

Estatística

$$p_{99} = 4,14$$

$$a) I. X \sim N(2,04; 0,6014)$$

$$\text{media} = 2,04 \quad \text{variancia} = \sigma^2 = 0,6014 \quad \text{desvio: } \sigma = 0,77$$

$$P(X \leq 2,81) \rightarrow Z = \frac{2,81 - 2,04}{0,77} = 0,97 \approx 0,99$$

$$P(X \leq 2,81) = 0,93044$$

$$II. P(X > 1,8) \quad Z > \frac{1,8 - 2,04}{0,77} = -0,308 \quad Z < 0,308 \approx 0,62103$$

$$III. P(1,01 \leq X \leq 2,5)$$

$$Z_1 = \frac{1,01 - 2,04}{0,77} = -1,3205$$

$$Z_2 = \frac{2,5 - 2,04}{0,77} = 0,5974$$

$$Z < 1,305 \approx 0,0939$$

$$Z < 0,5974 \approx 0,7224$$

$$P = 0,7224 - 0,0939 = 0,6290$$

$$B) P(2,20 \leq X \leq 3,20) \rightarrow Z_1 = \frac{2,20 - 2,04}{0,77} \approx 0,2051 \approx 0,57924$$

$$Z_2 = \frac{3,2 - 2,04}{0,77} = 1,5065 \approx 0,93774$$

$$P = 0,93776 - 0,57924 = 0,4085 \cdot 100 = 40,8$$

$$C) Z = 2,33$$

$$\text{LIM INF} = 2,04 - (2,33 \cdot 0,77) = 0,2226$$

$$\text{LIM SUP} = 2,04 + (2,33 \cdot 0,77) = 3,9574$$

$$\text{Intervalo} = [0,2226; 3,9574]$$

